

Notes: Broda, Limão, and Weinstein (AER, 2008)

Optimal Tariffs and Market Power: The Evidence

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1 One-Sentence Summary

This note summarizes Broda, Limão, and Weinstein (2008), which uses highly disaggregated trade, tariff, production, and elasticity estimates to test whether countries set higher tariffs on goods for which they face less elastic foreign export supply, showing that pre-WTO countries and unconstrained US trade barriers systematically reflect importer market power, even after accounting for revenue and lobbying motives.

2 Core Research Question

The paper asks whether the classic optimal-tariff logic is empirically relevant: do governments impose higher trade barriers on products where they have more market power in world markets? In the theory, a tariff has two opposing effects. It creates domestic distortions by reducing imports, but if foreign export supply is inelastic, it improves the importing country's terms of trade by lowering the price received by foreign exporters. The empirical prediction is therefore simple: absent international constraints, protection should be increasing in the inverse foreign export supply elasticity.

The authors focus on settings where trade policy is plausibly noncooperative. The main sample consists of fifteen countries before or around GATT/WTO accession, where tariffs were less constrained by reciprocal trade agreements. They then provide a second test using the United States, a WTO member, but only for trade barriers that are not tightly governed by WTO commitments, such as statutory tariffs and nontariff barriers.

3 Incremental Contribution

The paper's incremental contribution is substantial along four margins.

1. **It turns a century-old theory into a testable empirical object.** Optimal-tariff theory predicts that tariffs rise with importer market power, but earlier empirical trade-policy work did not directly test whether product-level tariffs respond to foreign export supply elasticities.
2. **It constructs product-country measures of market power.** The paper estimates importer-specific foreign export supply elasticities at the HS4 level for thousands of country-product observations using the Feenstra-Broda-Weinstein methodology.
3. **It separates market-power motives from political-economy motives.** By controlling for industry-by-country fixed effects and adding revenue and lobbying variables, the paper shows that the terms-of-trade motive is not merely a proxy for standard protection-for-sale forces.
4. **It links WTO constraints to the attenuation of market-power-based protection.** The US evidence shows that market power predicts unconstrained US barriers, but not WTO-bound MFN tariffs, supporting the view that trade agreements discipline terms-of-trade externalities.

4 Theoretical Framework

4.1 Benchmark optimal tariff

The paper begins from a standard large-country tariff model. Each country consumes a numeraire and nonnumeraire goods. For an imported good g , the domestic price is

$$p_g = (1 + \tau_g)p_g^*,$$

where p_g^* is the foreign export price and τ_g is the ad valorem tariff. Market clearing requires

$$m_g((1 + \tau_g)p_g^*) = m_g^*(p_g^*),$$

where m_g is home import demand and m_g^* is foreign export supply.

The welfare-maximizing tariff balances the domestic distortion against the terms-of-trade gain. If the country has no market power, foreign export supply is perfectly elastic and the optimal tariff is zero. If foreign export supply is upward sloping, the optimal tariff is positive and equals the inverse foreign export supply elasticity:

$$\tau_g^{opt} = \omega_g = \left[\left(\frac{dm_g^*}{dp_g^*} \right) \left(\frac{p_g^*}{m_g^*} \right) \right]^{-1}.$$

Thus the central prediction is:

$$\frac{\partial \tau_g}{\partial \omega_g} > 0.$$

Products with more inelastic foreign export supply should receive higher tariffs.

4.2 Political economy extension

The positive relationship also survives in a Grossman-Helpman style lobbying environment. In the large-country version of the model, the tariff contains both a terms-of-trade component and a political-economy component:

$$\tau_g^{GH} = \omega_g + \frac{I_g - \alpha_L}{a + \alpha_L} \frac{z_g}{\sigma_g},$$

where ω_g is inverse foreign export supply elasticity, z_g is inverse import penetration, σ_g is import demand elasticity, I_g indicates political organization, and a captures the government's weight on welfare. The key point is that even when governments care about lobbying, market power remains a first-order determinant of protection.

5 Data

5.1 Main cross-country sample

The main sample consists of fifteen countries with tariff data before GATT/WTO constraints: Algeria, Belarus, Bolivia, China, Czech Republic, Ecuador, Latvia, Lebanon, Lithuania, Oman, Paraguay, Russia, Saudi Arabia, Taiwan, and Ukraine. Tariff data come from TRAINS; trade data come mainly from COMTRADE; production data for lobbying controls come from UNIDO where available.

The unit of analysis is the importer-country by HS4 product. A “good” is an HS4 category, and a “variety” is an HS6-exporter pair within that HS4 category. This distinction matters because the elasticity estimation uses variation across varieties within a good.

5.2 Why pre-WTO data?

The theoretical prediction concerns tariffs set noncooperatively. Since the GATT/WTO is designed to internalize terms-of-trade externalities through reciprocal tariff reductions, WTO-bound tariffs may no longer reveal the unilateral optimal-tariff motive. The authors therefore use pre-WTO tariffs for the main test and later use US trade barriers that remain outside WTO constraints.

6 Estimating Market Power

6.1 Elasticity system

The authors estimate both import demand elasticity and inverse foreign export supply elasticity using the Feenstra (1994) methodology, extended by Broda and Weinstein (2006). The system is estimated at the importer-good level using variation across imported varieties:

$$\begin{aligned}\Delta^k \ln s_{igvt} &= -(\sigma_{ig} - 1)\Delta^k \ln p_{igvt} + \epsilon_{igvt}, \\ \Delta^k \ln p_{igvt} &= \frac{\omega_{ig}}{1 + \omega_{ig}}\Delta^k \ln s_{igvt} + \delta_{igvt}.\end{aligned}$$

Here s_{igvt} is the share of variety v of good g in importer i at time t , p_{igvt} is the domestic price or unit value, σ_{ig} is the import demand elasticity, and ω_{ig} is inverse foreign export supply elasticity. The differencing is both over time and relative to a benchmark variety within the same good.

The identifying assumption is that relative demand and supply shocks are uncorrelated. A key advantage of the double-differencing approach is that tariff changes common to all exporters of a good are differenced out, reducing reverse-causality concerns when elasticities are later used to explain tariffs.

6.2 Interpretation of inverse export supply elasticity

The market-power variable is ω_{ig} . A larger value means the importing country faces a less elastic foreign export supply curve for that product and therefore has more ability to push down exporter prices through tariffs. A low value means the country is closer to being a small buyer for that product.

7 Empirical Specification

The main tariff regression is

$$\tau_{ig} = \beta f(\omega_{ig}) + \eta_{iG} + x'_{ig}\gamma + u_{ig},$$

where τ_{ig} is the average ad valorem tariff at the HS4 level, ω_{ig} is market power, η_{iG} can include country effects, industry effects, or industry-by-country effects, and x_{ig} includes additional political economy or revenue controls.

The authors use three functional forms for market power:

- a linear inverse elasticity measure;
- a semi-log measure, $\log(1/\text{export elasticity}) = \log(\omega_{ig})$;
- a categorical dummy for goods with medium or high market power, defined relative to the within-country distribution of inverse elasticities.

Because the elasticity estimates can be noisy, the paper uses instrumental variables. The main instrument for a country's market power in a good is the average market power for that same good in the other countries. This exploits cross-country correlation in product-level supply conditions while reducing country-product-specific endogeneity and measurement-error concerns.

8 Main Findings

8.1 Elasticity estimates are economically plausible

The median inverse foreign export supply elasticity is around 1.6 across the full sample. The estimates pass several validation checks: they are correlated across countries for the same good; differentiated goods have much higher inverse elasticities than commodities; and larger or more remote countries tend to have more market power.

The product-type result is especially intuitive. Commodities have low estimated market power because suppliers can redirect homogeneous goods to alternative buyers. Differentiated products have higher market power because varieties are less substitutable and market segmentation is stronger.

8.2 Tariffs rise with market power

The paper finds a strong positive relationship between market power and tariffs. In IV specifications with industry-by-country fixed effects, goods with medium or high market power receive tariffs about 9 percentage points higher than low-market-power goods. This magnitude is close to the average tariff in the sample.

In the semi-log specification, the coefficient on market power is about 1.7 to 1.9 percentage points, implying that a one-log-point increase in inverse export supply elasticity raises tariffs by roughly two percentage points. These effects are substantially larger under IV than under OLS, consistent with attenuation from measurement error.

8.3 The result holds within countries

Country-by-country estimates are positive for every country and significant in most cases. The typical country coefficient is close to the pooled estimate, so the pooled finding is not driven only by China or by a few high-tariff countries. Larger countries tend to have larger absolute coefficients, but when scaled by mean tariff, the elasticity of tariffs with respect to market power is relatively stable across countries.

8.4 Revenue and lobbying do not explain away the result

The paper adds two important controls: inverse import demand elasticity to capture tariff-revenue motives, and the Grossman-Helpman lobbying variable, inverse import penetration divided by import demand elasticity. Market power remains large and significant. In the lobbying-augmented model, the market-power effect is at least as important as the lobbying effect and often larger in explaining tariff variation.

8.5 US unconstrained barriers also reflect market power

The US evidence provides an external validity test. The authors estimate US market power and examine two types of unconstrained trade barriers: nontariff barriers and statutory tariffs applied

to non-MFN countries. Both are significantly higher in products where the United States has more market power. By contrast, WTO-constrained MFN tariffs show a much weaker and statistically insignificant relationship with market power. This supports the interpretation that the WTO disciplines terms-of-trade-based protection.

9 Economic Magnitudes

The economic magnitude is large. The authors estimate that market-power motives raise tariffs by around 8 to 11 percentage points for medium/high-market-power goods. For the typical country, this is nearly as large as its average tariff. Since the effect applies to about two-thirds of tariff lines, removing this motive would imply tariff reductions of around 60 percent of existing levels.

The exporter-price implications are also meaningful. If tariffs driven by market power were removed, foreign exporters would receive higher prices, especially in large countries. For China, Russia, and Taiwan, the predicted exporter price gains are particularly large. Conversely, if a large country abandoned trade-agreement discipline and reasserted market-power-based tariffs, exporters would face substantial losses.

10 Identification and Robustness

The paper's identification strategy rests on several layers:

1. **Noncooperative policy environment:** tariffs are measured before WTO constraints or for US instruments not tightly constrained by WTO rules.
2. **Within-country-product variation:** the main tests compare products within countries, not only average tariffs across countries.
3. **Industry-by-country fixed effects:** these absorb broad political, sectoral, and institutional determinants of protection.
4. **Instrumental variables:** product-level market power in other countries instruments for domestic product-level market power, addressing measurement error and omitted country-product concerns.
5. **Alternative functional forms:** results hold using linear, log, and categorical measures of market power.
6. **Political economy controls:** results survive controls for tariff revenue and lobbying motives.
7. **US placebo-like comparison:** market power predicts unconstrained US barriers but not WTO-constrained MFN tariffs.

11 How to Read the Tables and Figures

11.1 Table 1. Data Sources and Years

Table 1 lists the fifteen countries, their GATT/WTO accession dates, production data sources and years, tariff data years, and trade data years. Its role is to establish that the main sample is designed around countries for which tariffs can be observed before WTO accession or before the relevant constraints are binding.

11.2 Table 2. Trade and Tariff Data Summary Statistics

Table 2 summarizes the scale of the data. It reports numbers of imported varieties, HS4 goods, median varieties per HS4 good, and tariff moments. The table shows substantial tariff variation both across countries and within countries. The last column indicates that most HS6 tariff variation occurs across HS4 categories, validating the HS4 aggregation used in the empirical design.

11.3 Figure 1. Tariff Distribution by Country

Figure 1 shows within-country distributions of HS4 tariffs. It makes clear that tariff schedules are not smooth continuous functions of any single determinant; many countries bunch tariffs at certain values or show truncation at zero. This motivates the paper’s flexible empirical approach using log and categorical market-power measures.

11.4 Tables 3A and 3B. Inverse Export Supply Elasticity Statistics

Table 3A reports summary statistics for inverse export supply elasticities. The median inverse elasticity is about 1.6 in the full sample, with clear separation between low, medium, and high market-power goods. Table 3B uses bootstrap intervals to show that the low-market-power category can be statistically distinguished from medium/high market-power categories.

11.5 Tables 4 and 5. Validating Elasticity Estimates

Table 4 shows that inverse export supply elasticities are positively correlated across countries for the same goods, suggesting that the estimates capture product-level supply conditions rather than pure noise. Table 5 shows that differentiated goods have much higher inverse elasticities than reference-priced goods or commodities, consistent with economic intuition.

11.6 Figure 2. Median Inverse Elasticities by Product Type

Figure 2 visualizes the Rauch classification result: differentiated products generally show the highest market power, commodities the lowest, and reference-priced goods lie between. This is one of the most intuitive checks on the estimated elasticities.

11.7 Table 6. Inverse Elasticities, GDP, Remoteness, and Import Shares

Table 6 confirms that larger countries, more remote countries, and countries with larger world import shares tend to face higher inverse export supply elasticities. This supports the idea that country size and market segmentation create market power, but also shows that product-level elasticities contain more information than GDP alone.

11.8 Figure 3. Median Tariffs and Market Power across Countries

Figure 3 provides a simple cross-country plot: countries with higher median inverse export supply elasticities tend to have higher median tariffs. This is suggestive but not conclusive, because average tariff levels may also reflect many country-level factors. The paper’s main evidence therefore comes from within-country regressions.

11.9 Tables 7 and 8. Baseline Tariff Regressions

Table 7 gives OLS and Tobit estimates. Market power has a positive effect on tariffs, but the linear effect is small due to skewness and outliers. Log and categorical measures show clearer positive effects. Table 8 presents IV estimates. The IV coefficients are much larger and economically important: medium/high-market-power goods receive tariffs about 9 percentage points higher than low-market-power goods.

11.10 Table 9. Country-by-Country IV Estimates

Table 9 shows that the market-power effect is positive in every country and statistically significant in most countries. China has the largest coefficient, consistent with its large size and high average tariff. The typical country coefficient is close to the pooled estimate.

11.11 Table 10. Market Power versus Tariff Revenue or Lobbying

Table 10 adds controls for revenue and lobbying motives. The market-power coefficient remains almost unchanged, demonstrating that the terms-of-trade motive is not absorbed by import-demand elasticity or standard lobbying variables.

11.12 Table 11. Market Power and Lobbying by Country

Table 11 repeats the lobbying-augmented specification for the seven countries with production data. Market power remains positive in each country, and its quantitative importance is comparable to or larger than the lobbying channel.

11.13 Table 12. Economic and Statistical Importance

Table 12 summarizes the magnitude. The market-power effect is about 8 to 11 percentage points in tariff levels and is close to the average tariff of the typical country. It also explains more tariff variation than the lobbying variable in the main comparison.

11.14 Table 13. US Trade Barriers

Table 13 shows that US nontariff barriers and statutory tariffs are significantly higher in goods where the United States has more market power. The relationship is weak for WTO-constrained MFN tariffs, reinforcing the interpretation that reciprocal trade agreements reduce the use of market-power-based protection.

12 Interpretation for Trade Policy

The paper supports the Bagwell-Staiger view that terms-of-trade externalities are central to the logic of trade agreements. If countries independently set tariffs, they use tariffs to exploit market power. Reciprocal trade agreements can reduce this incentive by internalizing the costs imposed on trading partners. The US evidence is especially important because it shows the same country behaves differently depending on whether the trade instrument is constrained by WTO rules.

13 Limitations and Caveats

Several caveats are useful when reading the paper. First, export supply elasticities are estimated over a relatively short horizon, so long-run elasticities may differ. Second, some tariff schedules are bunched or censored, which means simple linear optimal-tariff formulas are not expected to fit exactly. Third, the IV strategy is persuasive but relies on the assumption that market power in the same good in other countries affects domestic tariffs only through the domestic market-power variable and not through omitted product characteristics. Fourth, production data for lobbying controls are available only for a subset of countries, limiting some robustness tests.

14 Connection to Future Research

For research on tariffs, policy wedges, and firm responses, this paper is foundational because it treats tariffs not merely as exogenous trade costs but as endogenous policy choices shaped by market power. It suggests that product-level tariff exposure is not random: products facing inelastic foreign supply or differentiated supply conditions may be systematically targeted. Future work can build on this insight by studying dynamic policy adjustment, tariff exemptions, retaliation, firm sourcing responses, and how organizations reallocate capital, credit, or managerial attention across products with different exposure to endogenous trade policy.

15 Visual Appendix: Tables and Figures from the Paper

All items below are extracted directly from the uploaded paper. The interpretation for each table or figure is placed beside the corresponding visual so that the empirical message can be read together with the original evidence.

15.1 Table 1. Data Sources and Years

2040

THE AMERICAN ECONOMIC REVIEW

DECEMBER 2008

TABLE 1—DATA SOURCES AND YEARS

	GATT/WTO	Production data		Tariff data ^a	Trade data ^b
	Accession date	Source	Years		
Algeria				93	93–03
Belarus				97	98–03
Bolivia ^c	8-Sep-1990	UNIDO	93	93	93–03
China	11-Dec-2001	UNIDO	93	93	93–03
Czech ^d	15-Apr-1993			92	93–03
Ecuador	21-Jan-1996	UNIDO	93	93	94–03
Latvia	10-Feb-1999	UNIDO	96	97	94–03
Lebanon				00	97–02
Lithuania	31-May-2001	UNIDO	97	97	94–03
Oman	9-Nov-2000			92	94–03
Paraguay	6-Jan-1994			91	94–03
Russia				94	96–03
Saudi Arabia	11-Dec-2005			91	93–03
Taiwan	1-Jan-2002	UNIDO	96	96	92–96
Ukraine		UNIDO	97	97	96–02

Table 1. Data Sources and Years

Interpretation. Table 1 defines the empirical sample. The authors select fifteen countries for which tariff data can be observed before, or effectively before, GATT/WTO constraints bind. This matters because the theoretical prediction concerns *noncooperative* tariff setting: if a country is already constrained by reciprocal trade agreements, observed tariffs may no longer reveal the unilateral market-power motive. The table also shows the different data sources: TRAINS for tariffs, COMTRADE or TRAINS for trade, and UNIDO for production data in the subset where lobbying variables can be constructed.

15.2 Table 2. Trade and Tariff Data Summary Statistics

	Trade data			Tariff data				
	Number of varieties ^a	Number of HS4 goods	Median # of var. per HS4	Rate per four-digit HS			Fraction of HS6 variation between HS4	
				Observations ^b	Mean	Standard Deviation		
Algeria	26,466	1,100	13	739	23.8	17.4	15.6	0.95
Belarus	24,440	1,172	12	703	12.4	7.8	10.0	0.94
Bolivia	18,592	1,064	9	647	9.8	0.8	10.0	0.63
China	63,764	1,217	33	1,125	37.9	26.0	30.3	0.93
Czech Republic	61,781	1,219	30	1,075	9.5	17.6	5.1	0.87
Ecuador	22,979	1,101	11	753	9.8	5.5	10.6	0.91
Latvia	33,790	1,128	17	872	7.3	10.5	1.0	0.90
Lebanon	34,187	1,109	15	782	17.1	14.8	15.0	0.87
Lithuania	34,825	1,159	17	811	3.6	7.4	0.0	0.90
Oman	20,482	1,107	10	629	5.7	8.7	5.0	0.76
Paraguay	15,430	1,049	7	511	16.1	11.3	14.0	0.91
Russian	66,731	1,187	34	1,029	10.7	11.0	5.7	0.95
Saudi Arabia	62,525	1,202	32	1,036	12.1	2.6	12.0	0.93
Taiwan	38,397	1,215	19	891	9.7	8.5	7.5	0.90
Ukraine	37,693	1,128	18	730	7.4	7.6	5.0	0.95
Median	34,187	1,128	17	782	9.8	8.7	10.0	0.91

^a Varieties are defined as six-digit HS, exporting country pairs.

^b Number of observations for which elasticities and tariffs are available.

become imprecise. Therefore, in estimating (8) and (9) we define a good, g , as a four-digit HS cat-

Table 2. Trade and Tariff Data Summary Statistics

Interpretation. Table 2 shows that the paper has rich within-country product variation. A typical country has about 1,100 HS4 goods and many HS6-exporter varieties per HS4 category. Tariffs vary strongly across and within countries. The final column is especially important: most HS6 tariff variation occurs across HS4 goods rather than within HS4 goods, which supports the authors' decision to aggregate tariffs to the HS4 level, matching the level at which export supply elasticities are estimated.

15.3 Figure 1. Tariff Distribution by Country

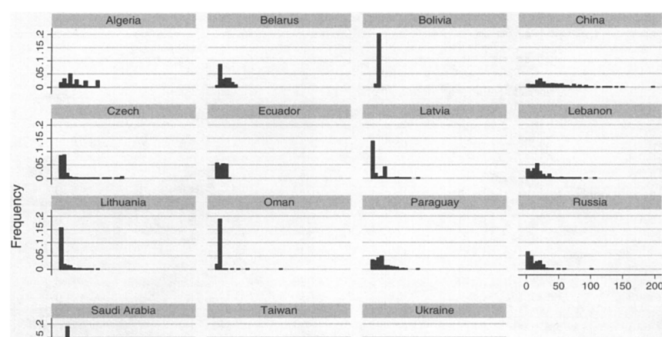


Figure 1. Tariff Distribution by Country

Interpretation. Figure 1 plots the distribution of HS4 tariffs within each sample country. The picture shows that tariffs are often bunched, censored at zero, or concentrated at a few policy-relevant rates. This is why the paper does not rely only on the exact linear optimal-tariff formula. Instead, it uses flexible market-power measures, including log inverse elasticities and medium/high-market-power dummies.

15.4 Table 3A. Inverse Export Supply Elasticity Statistics

TABLE 3A—INVERSE EXPORT SUPPLY ELASTICITY STATISTICS

Statistic	Observations ^a	Median ^b			Mean		Standard deviation	
Sample	All	Low	Medium	High	All	W/out top decile	All	W/out top decile
Algeria	739	0.4	2.8	91	118	23	333	47
Belarus	703	0.3	1.5	61	85	15	257	36
Bolivia	647	0.3	2.0	91	102	23	283	49
China	1,125	0.4	2.1	80	92	17	267	35
Czech Republic	1,075	0.3	1.4	26	63	7	233	18
Ecuador	753	0.3	1.5	56	76	13	243	30
Latvia	872	0.2	1.1	9	52	3	239	8
Lebanon	782	0.1	0.9	31	56	7	215	18
Lithuania	811	0.3	1.2	24	65	6	235	16
Oman	629	0.3	1.2	25	209	7	3,336	21
Paraguay	511	0.4	3.0	153	132	67	315	169
Russia	1,029	0.5	1.8	33	48	8	198	18
Saudi Arabia	1,036	0.4	1.7	50	71	11	232	25
Taiwan	891	0.1	1.4	131	90	20	241	43
Ukraine	730	0.4	2.1	78	86	16	254	34
Median	782	0.3	1.6	54	85	13	243	30

^aNumber of observations for which elasticities and tariffs are available. The tariff availability did not bind except for Ukraine, where it was not available for about 130 HS4 goods for which elasticities were computed.

^bThe median over the “low” sample corresponds to the median over the bottom tercile of inverse elasticities. Medium and high correspond to the second and third terciles.

Table 3A. Inverse Export Supply Elasticity Statistics

Interpretation. Table 3A summarizes the estimated inverse foreign export supply elasticities, the paper’s market-power measure. The median inverse elasticity is about 1.6 in the pooled sample. Low-market-power goods have much smaller inverse elasticities than medium and high market-power goods, meaning that the categorical distinction used later in the tariff regressions is economically meaningful. The table also shows why the authors worry about outliers: means and standard deviations are much larger when the top decile is included.

15.5 Table 3B. Bootstrapped Statistics for Inverse Export Supply Elasticities

TABLE 3B—BOOTSTRAPPED STATISTICS FOR INVERSE EXPORT SUPPLY ELASTICITIES

Statistic Sample	Low		Medium or high	
	Median	Confidence interval ^a	Median	Confidence interval ^a
Algeria	0.5	[0.10 , 0.8]	5.0	[2.0 , 81]
Belarus	0.3	[0.03 , 0.5]	3.0	[0.9 , 58]
Bolivia	0.4	[0.02 , 0.6]	4.2	[1.1 , 87]
China	0.6	[0.15 , 0.8]	5.0	[1.5 , 59]
Czech Republic	0.3	[0.06 , 0.5]	3.0	[0.9 , 30]
Ecuador	0.4	[0.02 , 0.6]	3.3	[0.9 , 63]
Latvia	0.3	[0.02 , 0.4]	2.3	[0.7 , 60]
Lebanon	0.2	[0.01 , 0.3]	2.1	[0.6 , 29]
Lithuania	0.3	[0.03 , 0.5]	2.3	[0.7 , 28]
Oman	0.4	[0.04 , 0.6]	2.2	[0.6 , 35]
Paraguay	0.5	[0.03 , 0.8]	6.7	[1.9 , 98]
Russia	0.6	[0.12 , 0.7]	3.8	[1.3 , 42]
Saudi Arabia	0.5	[0.10 , 0.7]	4.1	[1.4 , 44]
Taiwan	0.3	[0.01 , 0.3]	3.0	[0.8 , 98]
Ukraine	0.6	[0.08 , 0.9]	4.5	[1.4 , 59]
Median	0.4	[0.04 , 0.6]	3.4	[1.1 , 49]

Notes: “Median” indicates the median of the 250 bootstrapped estimates for each inverse elasticity, ω_{ig} . The “low” column reports the median of that value in the bottom tercile of the sample for ω_{ig} in a country. The value in the medium or high column corresponds to the median in the rest of the sample.

^a The lower bound of the confidence interval reported is the median lower bound over all those estimated in the relevant part of the sample; similarly for the upper bound. The individual estimates for the 1 – 7% confidence interval are

Table 3B. Bootstrapped Statistics for Inverse Export Supply Elasticities

Interpretation. Table 3B addresses precision. The bootstrapped confidence intervals show that the low-market-power group is generally distinguishable from the medium/high-market-power group. This supports the later IV strategy, where the authors instrument continuous market power with information based on cross-country product-level market-power rankings.

15.6 Table 4. Correlation of Inverse Export Supply Elasticities across Countries

TABLE 4—CORRELATION OF INVERSE EXPORT SUPPLY ELASTICITIES ACROSS COUNTRIES

Dependent variable: Statistic	Log inverse export supply			
	Beta	Standard error	R ²	Number of observations
Algeria	0.80	(0.07)	0.13	739
Belarus	0.80	(0.07)	0.14	703
Bolivia	0.82	(0.09)	0.13	647
China	0.54	(0.06)	0.11	1,125
Czech Republic	0.61	(0.05)	0.12	1,075
Ecuador	0.73	(0.08)	0.12	753
Latvia	0.57	(0.07)	0.09	872
Lebanon	0.71	(0.08)	0.11	782
Lithuania	0.70	(0.07)	0.13	811
Oman	0.39	(0.08)	0.04	629
Paraguay	0.94	(0.11)	0.14	511
Russia	0.53	(0.05)	0.11	1,029
Saudi Arabia	0.48	(0.06)	0.08	1,036
Taiwan	0.31	(0.08)	0.02	891
Ukraine	0.83	(0.07)	0.17	730
Median	0.70	(0.07)	0.12	782

Note: Univariate regression of log inverse export supply elasticities in each country on the average of the log inverse elasticities in that good for the remaining 14 countries.

Table 4. Correlation of Inverse Export Supply Elasticities across Countries

Interpretation. Table 4 validates the elasticity estimates by showing that inverse elasticities for the same HS4 goods are positively correlated across countries. This suggests that the estimates capture real product-level supply conditions rather than only importer-specific noise. This cross-country correlation is also the basis for the paper’s instrument: market power for the same good in other countries predicts domestic market power.

15.7 Table 5. Inverse Elasticities by Product Type

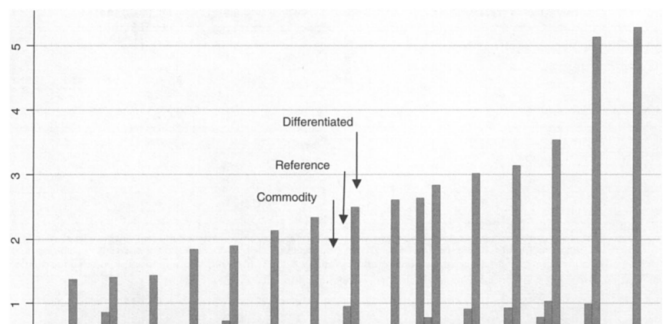
TABLE 5—INVERSE ELASTICITIES BY PRODUCT TYPE			
	Differentiated	Reference priced	Commodity
Median inv elasticity	2.38	0.70	0.45
Standard errors	(0.04)	(0.06)	(0.14)
p-value: Differentiated vs. refer. or commod.	0.00	0.00	0.00
Mean inv. elasticity	17.5	9.3	8.3
Standard errors	(0.71)	(0.70)	(1.23)
p-value: Differentiated vs. refer. or commod.	0.00	0.00	0.00

Notes: The number of observations for the median regression is 8,734, less than the full sample since not all HS4 can be uniquely matched to Rauch's classification. The number for the mean regression is 7,927 because we trim the top

Table 5. Inverse Elasticities by Product Type

Interpretation. Table 5 uses Rauch's product classification to check whether the elasticities make economic sense. Differentiated products have much higher inverse export supply elasticities than reference-priced goods or commodities. This is exactly what theory suggests: when products are differentiated, foreign exporters have fewer outside markets and importers have more ability to affect exporter prices.

15.8 Figure 2. Median Inverse Elasticities by Product Type



15.9 Table 6. Inverse Export Supply Elasticities, GDP, Remoteness, and Import Shares

TABLE 6—INVERSE EXPORT SUPPLY ELASTICITIES, GDP, REMOTENESS, AND IMPORT SHARES			
Dependent variable	Log inverse export supply		
Log GDP	0.17 (0.04)	0.18 (0.03)	
Log remoteness		0.40 (0.15)	
Share of world HS4 imports			7.19 (1.48)
Observations	12,343	12,343	12,343
R^2	0.26	0.26	0.25
R^2 within	0.01	0.02	0.00

Table 6. Inverse Export Supply Elasticities, GDP, Remoteness, and Import Shares

Interpretation. Table 6 links market power to country characteristics. Larger countries, more remote countries, and countries with larger import shares tend to face higher inverse export supply elasticities. The result is intuitive: a large or geographically segmented buyer is harder for exporters to replace, so its tariff can move exporter prices more strongly.

15.10 Figure 3. Median Tariffs and Market Power across Countries

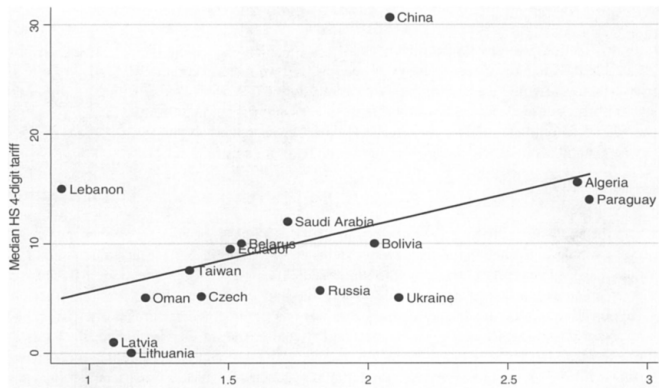


Figure 3. Median Tariffs and Market Power across Countries

Interpretation. Figure 3 provides a simple cross-country preview of the main result. Countries with higher median market power tend to have higher median tariffs. The figure is only suggestive because country-level tariff averages may reflect many omitted factors, but it motivates the more credible within-country, within-industry regressions that follow.

15.11 Table 7. Tariffs and Market Power across Goods: OLS and Tobit Estimates

TABLE 7—TARIFFS AND MARKET POWER ACROSS GOODS (WITHIN COUNTRIES): OLS AND TOBIT ESTIMATES									
Dependent variable	Average tariff at four-digit HS (%)								
Fixed effects	Country			Country and industry					
Estimation method	OLS	OLS	OLS	OLS	OLS	OLS	Tobit	OLS ^a	OLS
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Inverse exp. elast.	0.0003 (0.0001)			0.0004 (0.0004)					
Mid and high inv exp elast		1.24 (0.25)			1.46 (0.24)			1.86 (0.31)	
Log(1/export elasticity)			0.12 (0.04)			0.17 (0.04)	0.17 (0.05)		
(Inv. exp. elast) × (1 – med hi)								1.45 (0.31)	
(Inv. exp. elast) × med hi								0.0003 (0.0001)	
Mid inv. exp. elast.									1.56 (0.28)
High inv. exp. elast.									1.37 (0.28)
Algeria	23.8 (0.64)	23.0 (0.65)	23.6 (0.64)	24.6 (0.95)	23.6 (0.96)	24.3 (0.95)	24.3 (0.93)	23.1 (0.97)	23.6 (0.96)
Belarus	12.3 (0.29)	11.5 (0.33)	12.2 (0.29)	12.6 (0.76)	11.6 (0.78)	12.5 (0.76)	12.4 (0.94)	11.3 (0.79)	11.7 (0.78)
Bolivia	9.8 (0.03)	9.0 (0.17)	9.7 (0.06)	10.1 (0.73)	9.2 (0.75)	10.0 (0.73)	10.0 (0.95)	8.8 (0.77)	9.2 (0.75)
China	37.8 (0.77)	37.0 (0.79)	37.7 (0.77)	38.2 (0.98)	37.2 (1.01)	38.0 (0.99)	37.9 (0.89)	36.6 (1.03)	37.2 (1.01)
Czech Republic	9.5 (0.53)	8.7 (0.53)	9.4 (0.53)	9.7 (0.85)	8.7 (0.86)	9.6 (0.85)	8.8 (0.89)	8.3 (0.87)	8.7 (0.86)
Ecuador	9.8 (0.19)	9.0 (0.26)	9.7 (0.20)	10.3 (0.73)	9.4 (0.74)	10.2 (0.73)	10.1 (0.93)	9.0 (0.76)	9.4 (0.74)
Latvia	7.3 (0.35)	6.4 (0.40)	7.2 (0.35)	7.3 (0.76)	6.3 (0.78)	7.2 (0.76)	6.9 (0.91)	6.0 (0.79)	6.3 (0.78)
Lebanon	17.1 (0.53)	16.2 (0.56)	17.0 (0.53)	17.1 (0.84)	16.1 (0.86)	17.0 (0.84)	17.0 (0.92)	15.9 (0.86)	16.1 (0.86)
Lithuania	3.6 (0.26)	2.8 (0.31)	3.6 (0.26)	3.6 (0.74)	2.6 (0.76)	3.5 (0.74)	–6.0 (0.98)	2.3 (0.77)	2.6 (0.76)
Oman	5.6 (0.34)	4.9 (0.37)	5.6 (0.34)	5.7 (0.77)	4.8 (0.79)	5.6 (0.77)	4.9 (0.94)	4.4 (0.79)	4.8 (0.79)
Paraguay	16.0 (0.49)	15.3 (0.52)	15.9 (0.50)	16.3 (0.84)	15.4 (0.85)	16.1 (0.84)	15.9 (0.99)	14.9 (0.86)	15.4 (0.85)
Russia	10.6 (0.34)	9.8 (0.38)	10.5 (0.34)	10.8 (0.77)	9.9 (0.79)	10.7 (0.77)	10.0 (0.89)	9.4 (0.82)	9.9 (0.79)
Saudi Arabia	12.1 (0.08)	11.3 (0.18)	12.0 (0.09)	12.4 (0.71)	11.4 (0.74)	12.2 (0.72)	12.1 (0.89)	10.9 (0.76)	11.4 (0.74)
Taiwan	9.7 (0.28)	8.9 (0.33)	9.6 (0.28)	10.3 (0.74)	9.3 (0.76)	10.1 (0.75)	9.7 (0.91)	9.0 (0.77)	9.3 (0.76)
Ukraine	7.4 (0.28)	6.6 (0.33)	7.2 (0.29)	8.1 (0.74)	7.1 (0.76)	7.9 (0.74)	6.8 (0.93)	6.6 (0.78)	7.1 (0.76)
Observations	12,333	12,333	12,333	12,333	12,333	12,333	12,333	12,333	12,333
Number of parameters	16	16	16	36	35	36	35	38	36
Adj. R ²	0.61	0.61	0.61	0.66	0.66	0.66			0.66

Notes: Standard errors in parentheses (all heteroskedasticity robust except Tobit). Industry dummies defined by section according to Harmonized Standard tariff schedule.

^aOptimal threshold regression based on minimum RSS found using a grid search over 50 points of the distribution of inverse exp. elast. (from first to ninety-ninth percentile in intervals of two). Optimal threshold is fifty-third percentile. Accordingly, med hi equals one above the fifty-third percentile and zero otherwise. Bruce E. Hansen (2000) shows that the dependence of the parameters on the threshold estimate is not of “first-order” asymptotic importance, so inference on them can be done as if the threshold estimate were the true value.

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Table 7. Tariffs and Market Power across Goods: OLS and Tobit Estimates

Interpretation. Table 7 gives baseline OLS and Tobit evidence. Across specifications, market power is positively related to tariffs. The log and dummy specifications are more informative than the raw linear inverse elasticity because the inverse elasticity is highly skewed and the marginal tariff response is unlikely to remain linear at very high market-power values. The estimates show the relationship survives country and industry controls.

15.12 Table 8. Tariffs and Market Power across Goods: IV Estimates

TABLE 8—TARIFFS AND MARKET POWER ACROSS GOODS (WITHIN COUNTRIES): IV ESTIMATES

Dependent variable	Average tariff at four-digit HS (%)								
Fixed effects	Country			Country and industry			Industry by country		
Estimation method	IV GMM (1)	IV GMM (2)	IV GMM (3)	IV GMM (4)	IV GMM (5)	IV GMM (6)	IV GMM (7)	IV GMM (8)	IV GMM (9)
Inverse exp. elast.	0.040 (0.027)			0.089 (0.055)			0.075 (0.028)		
Mid and high inv. exp. elast.		3.96 (0.76)			8.88 (1.18)			9.07 (1.08)	
Log(1/export elasticity)			0.75 (0.15)			1.71 (0.23)			1.73 (0.21)
Observations	12,258	12,258	12,258	12,258	12,258	12,258	12,258	12,258	12,258
No. of parameters	16	16	16	35	35	35	284	282	283
1st stage F	5	1649	1335	2	653	517	3	691	544

Notes: Standard errors in parentheses (heteroskedasticity robust). Industry dummies defined by section according to the Harmonized Standard tariff schedule.

example, the coefficient is 1.7 when we control for industry or industry-by-country effects in columns (6) and (9), respectively. This estimate is ten times larger than the OLS one and significant

Table 8. Tariffs and Market Power across Goods: IV Estimates

Interpretation. Table 8 is the core evidence. After instrumenting market power with product-level market power in other countries and controlling for industry-by-country effects, the market-power effect becomes large and significant. Goods with medium or high market power have tariffs about 9 percentage points higher than low-market-power goods. This is the paper's key empirical result.

15.13 Table 9. Tariffs and Market Power across Goods by Country: IV Estimates

TABLE 9A—TARIFFS AND MARKET POWER ACROSS GOODS BY COUNTRY: IV ESTIMATES										
Dependent variable	Average tariff at four-digit HS (%)									
Fixed effects	Ind. by Country	Ind. by Country	Ind.	Ind.	Ind.	Ind.	Ind.	Ind.	Ind.	Ind.
Estimation method	IV GMM	IV GMM	IV GMM	IV GMM	IV GMM	IV Tobit	IV GMM	IV GMM	IV GMM	IV GMM
Sample	All	Exc. Bol., Om., Sau.	China	Russia	Taiwan	Saudi Arabia	Ukraine	Czech.	Algeria	Belarus
Log(1/export elasticity)	1.73 (0.21)	2.11 (0.25)	7.60 (1.78)	2.42 (0.61)	1.98 (0.78)	1.75 (2.55)	0.71 (0.26)	0.16 (0.24)	5.40 (0.97)	2.28 (0.52)
Observations	12,258	9,952	1,089	1,021	841	1,031	685	1,000	739	703
No. of parameters	283	227	20	18	20	18	20	18	18	18
First stage F	544	477	45	44	12	7.6 ^a	48	61	60	40
Mean tariff (%)	13.4	14.2	38.2	10.3	8.9	12.2	5.8	5.5	23.8	12.4
Elasticity (at mean)	0.13	0.15	0.20	0.23	0.22	0.14	0.12	0.03	0.23	0.18

TABLE 9B—TARIFFS AND MARKET POWER ACROSS GOODS BY COUNTRY: IV ESTIMATES										
Dependent variable	Average tariff at four-digit HS (%)									
Fixed effects	Ind.	Ind.	Ind.	Ind.	Ind.	Ind.	Ind.	Ind.	Ind.	Ind.
Estimation method	IV GMM	IV Tobit	IV GMM	IV GMM	IV GMM	IV Tobit	IV GMM	IV Tobit	IV GMM	IV GMM
Sample	Ecuador	Oman	Paraguay	Lithuania	Lebanon	Bolivia	Latvia	Mean	Median	
Log(1/export elasticity)	1.55 (0.34)	0.60 (0.18)	2.44 (0.67)	0.83 (0.27)	2.41 (0.54)	0.79 (0.36)	1.41 (0.60)	2.15	1.75	
Observations	753	628	510	768	754	647	868			
No. of parameters	19	17	19	20	20	19	19			
First stage F	45	3.7 ^a	33	35	52	8.96 ^a	37			
Mean tariff (%)	9.8	5.7	16.0	2.3	15.0	9.8	7.0			

Table 9. Tariffs and Market Power across Goods by Country: IV Estimates

Interpretation. Table 9 shows that the main result is not driven by one country. The coefficient on log inverse export elasticity is positive in every country and statistically significant in most of them. China has a particularly large coefficient, consistent with its size and high average tariff. The median country estimate is close to the pooled estimate, suggesting the pooled result is representative.

15.14 Table 10. Market Power versus Tariff Revenue or Lobbying as a Source of Protection

TABLE 10—MARKET POWER VERSUS TARIFF REVENUE OR LOBBYING AS A SOURCE OF PROTECTION

Dependent variable	Average tariff at four-digit HS (%)					
Fixed effects	Industry by country					
Estimation method	IV GMM					
Sample	Pooled (all)		Pooled (all)		Pooled (7)	
Theory	Market power		Market power and tariff revenue		Market power and lobbying	
Mid and high inv. exp. elast.	9.07		9.04		10.20	
	(1.08)		(1.24)		(1.79)	
Mid and high inv. imp. elast.			−0.20			
			(2.08)			
Mid and hi inv. imp. pen/imp. elast.					6.28	
					(1.97)	
Log(1/export elasticity)	1.73		1.81		1.94	
	(0.21)		(0.23)		(0.38)	
Log(1/import elasticity)			−0.90			
			(0.81)			
Log(inv. imp. pen/imp. elas.)					1.59	
					(0.55)	
Observations	12,258	12,258	12,258	12,258	5,178	5,178
No. of parameters	282	283	283	284	132	133
First stage F (market power)	691	544	370	312	171	129
First stage F (other)	na	na	102	144	131	188

Notes: Standard errors in parentheses (heteroskedasticity robust). Industry dummies defined by section according to the Harmonized Standard tariff schedule. The countries with available data for the lobbying specifications are Bolivia, China, Ecuador, Latvia, Lithuania, Taiwan, and Ukraine. These data are not available for mining and agricultural

Table 10. Market Power versus Tariff Revenue or Lobbying as a Source of Protection

Interpretation. Table 10 asks whether the market-power result is actually capturing tariff revenue motives or lobbying. It is not. After adding inverse import demand elasticity and a Grossman-Helpman-style lobbying variable, the market-power coefficient remains very close to the baseline. This is central for the paper's contribution because it separates terms-of-trade motives from standard political-economy explanations.

15.15 Table 11. Market Power and Lobbying: IV Estimates by Country

TABLE 11—MARKET POWER AND LOBBYING: IV ESTIMATES BY COUNTRY										
<i>Panel A: Semilog</i>										
Dependent variable	Average tariff at four-digit HS (%)									
Fixed effects	Ind. by country	Ind.	Ind.	Ind.	Ind.	Ind.	Ind.	Ind.	Ind.	
Estimation method	IV GMM	IV GMM	IV GMM	IV GMM	IV GMM	IV GMM	IV Tobit	IV GMM		
Sample	Pooled (7)	China	Taiwan	Ukraine	Ecuador	Lithuania	Bolivia	Latvia	Mean	Median
Log(1/export elasticity)	1.94 (0.38)	4.69 (2.12)	2.39 (1.32)	0.91 (0.25)	1.81 (0.45)	0.84 (0.27)	0.97 (0.16)	1.52 (0.67)	1.87	1.5
Log(inv. imp. pen./imp. elas.)	1.59 (0.55)	6.21 (4.31)	0.43 (1.18)	0.97 (0.75)	0.27 (0.57)	1.64 (0.40)	0.21 (0.19)	1.89 (1.33)	1.66	1.0
Observations	5,178	861	780	616	712	706	618	788		
No. of parameters	133	21	20	19	20	20		20		
1st stage F : log(1/exp. el)	129	39	6	25	24	18	9	18		
1st st. F : log(imp. pen/imp. el)	188	37	32	7	47.2	28.5	18.4	24.9		
Mean tariff (%)	12.8	37.0	8.8	5.7	10.0	2.4	9.8	6.9		
Elasticity (at mean)	0.15	0.13	0.27	0.16	0.18	0.35	0.10	0.22	0.20	0.18
<i>Panel B: Dummy</i>										
Dependent variable	Average tariff at four-digit HS (%)									
Fixed effects	Ind. by country	Ind.	Ind.	Ind.	Ind.	Ind.	Ind.	Ind.	Ind.	
Estimation method	IV GMM	IV GMM	IV GMM	IV GMM	IV GMM	IV GMM	IV Tobit	IV GMM		
Sample	Pooled (7)	China	Taiwan	Ukraine	Ecuador	Lithuania	Bolivia	Latvia	Mean	Median
Mid and high inv. exp. elast.	10.2 (1.79)	22.9 (9.18)	13.3 (4.61)	4.2 (1.17)	10.3 (2.87)	3.4 (1.43)	8.0 (0.92)	6.7 (2.86)	9.83	8.0
Mid and high inv. imp. pen./imp. elast	6.28 (1.97)	16.1 (10.07)	0.9 (3.52)	1.4 (1.84)	1.5 (3.35)	6.4 (1.84)	2.5 (1.36)	6.4 (3.54)	5.01	2.5
Observations	5,178	861	780	616	712	706	618	788		
No. of parameters	132	21	20	19	20	20		20		
1st-stage F : log(1/exp. el)	171	48	10	36	27	24	9	29		
1st-stage F : log(imp. pen/imp. el)	131	37	19	11	23.9	14.0	17.8	19.7		
Mean tariff (%)	12.8	37.0	8.8	5.7	10.0	2.4	9.8	6.9		
Mid-hi inv. exp. elast. /mean (%)	80	62	151	73	103	140	82	97	101	97

Notes: Standard errors in parentheses (heteroskedasticity robust). Constant and industry dummies included but not reported. Industry dummies defined by section according to the Harmonized Standard tariff schedule. Observations with extreme outliers in terms of tariffs were dropped. This affects the sample for Taiwan, Lithuania, and Ukraine, but only slightly since the criteria drops fewer than 36 observations in any of these countries (those with tariff values more than three times the interquartile range above the seventy-fifth percentile or below the twenty-fifth). Bolivia is estimated via an IV Tobit procedure to account for the fact that a large fraction of their observations are censored from above and/or below. Given the lack of variation in its tariffs within industries, the estimation for Bolivia does not include industry dummies. The first-stage statistic in this case refers to the z -statistic of the relevant instrument.

Table 11. Market Power and Lobbying: IV Estimates by Country

Interpretation. Table 11 repeats the lobbying-augmented analysis country by country for the seven countries with production data. Market power remains positive across countries and is often comparable to or larger than the lobbying channel. The table therefore shows that the terms-of-trade motive is not only a pooled-sample artifact.

15.16 Table 12. Economic and Statistical Importance of Market Power in Tariff Setting

TABLE 12—ECONOMIC AND STATISTICAL IMPORTANCE OF MARKET POWER IN TARIFF SETTING: SUMMARY MEASURES

Variable	Measure	Specification				
		Market power			Market power and lobbying	
		Pooled (Table 8)	Subsample ^a (Table 8)	Typical country ^b (Table 9)	Pooled (Table 10)	Typical country (Table 11)
Log(1/export elasticity)	β	1.7 pp	2.1 pp	1.8 pp	1.9 pp	1.5 pp
	Elasticity ($\beta/\text{mean tariff}$)	0.13	0.15	0.16	0.15	0.18
	$\beta \times \text{s.d.}$	5 pp	6 pp	5 pp	5 pp	4 pp
	Elasticity relative to PE (β/γ)	.	.	.	1.2	0.9
	Impact relative to PE ($\beta \times \text{s.d.} / (\ln u) / \gamma \times \text{s.d.} / (\ln \tau / \sigma)$)	.	.	.	1.6	1.5

Table 12. Economic and Statistical Importance of Market Power in Tariff Setting

Interpretation. Table 12 translates coefficients into economic magnitudes. The market-power effect is about 8 to 11 percentage points, close to the average tariff of the typical country. It is also at least as important as the lobbying variable in explaining tariff variation. This is why the paper concludes that market power is not just statistically significant but quantitatively central.

15.17 Table 13. Market Power and Lobbying as a Source of Protection in the US

Panel A: Nontariff barriers								
Theory	Market power				Market power and lobbying			
Fixed effects	Industry				Industry			
Estimation method	IV Tobit				IV Tobit ^a			
Dependent variable	Coverage ratio (HS4) ^a		Advalorem equiv. (HS4, %)		Coverage ratio (HS4)		Advalorem equiv. (HS4, %)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Mid and high inv. exp. elast.	0.90		38.8		4.93		70.8	
	(0.31)		(15.73)		(1.52)		(21.99)	
Mid and hi inv. imp. pen./imp. elast					-0.08		3.99	
					(0.86)		(13.14)	
Log(1/export elasticity)		0.22		9.71		1.16		16.0
		(0.08)		(4.00)		(0.39)		(5.47)
Log(inv. imp. pen./imp. elas.)						0.19		4.74
						(0.34)		(4.94)
Observations ^c	804	804	804	804	708	708	708	708
Number of parameters	17	17	17	17	17	17	17	17
First stage z-stat (market power)	7.1	6.6	7.1	6.6	6.2	5.3	6.2	5.3
First stage z-stat (other)	na	na	na	na	10.1	11.4	10.1	11.4
Panel B: Tariff barriers								
Theory	Market power				Market power and lobbying			
Fixed effects	Industry				Industry			
Estimation method	IV Tobit				IV Tobit ^b			
Dependent variable	Non-WTO (HS4, %)		WTO (HS4, %)		Non-WTO (HS4, %)		WTO (HS4, %)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Mid and high inv. exp. elast.	21.2		1.52		26.9		1.89	
	(5.53)		(1.18)		(8.05)		(1.58)	
Mid and hi inv. imp. pen./imp. elast					10.8		-0.63	
					(4.91)		(0.96)	
Log(1/export elasticity)		5.07		0.36		5.58		0.45
		(1.36)		(0.28)		(1.86)		(0.38)
Log(inv. imp. pen./imp. elas.)						4.76		-0.18
						(1.69)		(0.34)
Observations ^c	870	870	869	869	775	775	774	774
Number of parameters	20	20	20	20	21	21	21	21
First stage z-stat (market power)	7.3	7.1	7.3	7.1	6.0	5.3	6.0	5.3
First stage z-stat (other)	na	na	na	na	10.0	11.6	10.0	11.6
Mean	30.6	30.6	3.4	3.4	33.0	33.0	3.7	3.7
Mid-hi inv. exp. elast. /mean (%)	69		45		81		51	
Elasticity (at mean)		0.17		0.11		0.17		0.12

Notes: Standard errors in parentheses. Industry dummies defined by section according to the Harmonized Standard tariff schedule.

^aCoverage ratio is defined as the fraction of HS6 tariff lines in a given HS4 category that had an NTB. Since it varies

Table 13. Market Power and Lobbying as a Source of Protection in the US

Interpretation. Table 13 extends the logic to the United States. Market power strongly predicts US nontariff barriers and statutory tariffs, both of which are less constrained by WTO rules. By contrast, the relationship is weak for WTO-constrained MFN tariffs. This contrast supports the interpretation that trade agreements reduce the use of market-power-based protection.

16 Bottom Line

The paper provides direct evidence that countries do not set tariffs as if they were price takers. When trade policy is unconstrained, tariffs are systematically higher in products where the importer faces inelastic foreign export supply. This market-power motive is economically large, robust to revenue and lobbying controls, and muted when WTO constraints bind.